

Factorial Design

7 Facets fully crossed:
Item (p) · Model (M) · Precision (π)
Prompt (δ) · Seed (s) · Temp (τ) · Ordering (o)

8 models × 5 benchmarks
MC: MMLU, ARC, HellaSwag
FF: GSM8K, MATH

→ **2.3M binary records**

G-Study

Variance Decomposition

Henderson Method I (ANOVA)
 $\sigma_{\text{item}}^2, \sigma_{\text{model} \times \text{item}}^2, \sigma_{\text{prompt} \times \text{item}}^2, \dots$

Key: model × item up to 36%
*Models differ in which items
are hard, not overall ability*

D-Study

Budget Projections

$$G(n) = \frac{n \cdot \sigma_p^2}{n \cdot \sigma_p^2 + \sigma_{\text{rel}}^2}$$

Target: $G \geq 0.80$

Single-condition: 552 items
3-prompt × 3-seed: 178 items
Full multi-facet: 55 items

3.1 × item reduction

C1 Reproducibility Budgets

Design-dependent item requirements

IRT + G-theory compose:
116 IRT items → $G = 0.457$ (single)
116 IRT items → $G = 0.895$ (multi-facet)

C2 Format-Associated Patterns (exploratory)

MC benchmarks: 68–72% item-related
FF benchmarks: 31–41% item-related
Bootstrap CIs non-overlapping (21pp gap)